waveform of the voltage is generated. For one full rotation of $\overrightarrow{OA}$, one cycle of voltage is produced.

The alternating current in the same circuit may be given by the relation $i = I_m \sin(\omega t + \phi)$ which is represented by another phasor $\overrightarrow{OB}$. Here ϕ is the phase angle between voltage and current. In this case, the current leads the voltage by phase angle ϕ which is shown in Figure 4.44. If the current lags behind the voltage, then we write $i = I_m \sin(\omega t - \phi)$.

4.7.3 AC circuit containing pure resistor

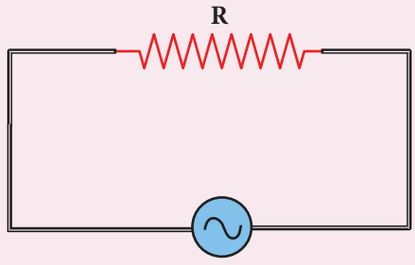


Figure 4.45 AC circuit with resistance

Consider a circuit containing a pure resistor of resistance R connected across an alternating voltage source (Figure 4.45). The instantaneous value of the alternating voltage is given by

$$\upsilon = V_m \sin \omega t \qquad (4.43)$$

An alternating current i flowing in the circuit due to this voltage, develops a potential drop across R and is given by

$$V_R = iR \qquad (4.44)$$

Kirchoff's loop rule (Refer section 2.4) states that the algebraic sum of potential differences in a closed circuit is zero. For this resistive circuit,

$$\upsilon - V_R = 0$$

From equation (4.43) and (4.44),

$$V_m \sin \omega t = iR$$

$$i = \frac{V_m}{R} \sin \omega t$$

$$i = I_m \sin \omega t \qquad (4.45)$$

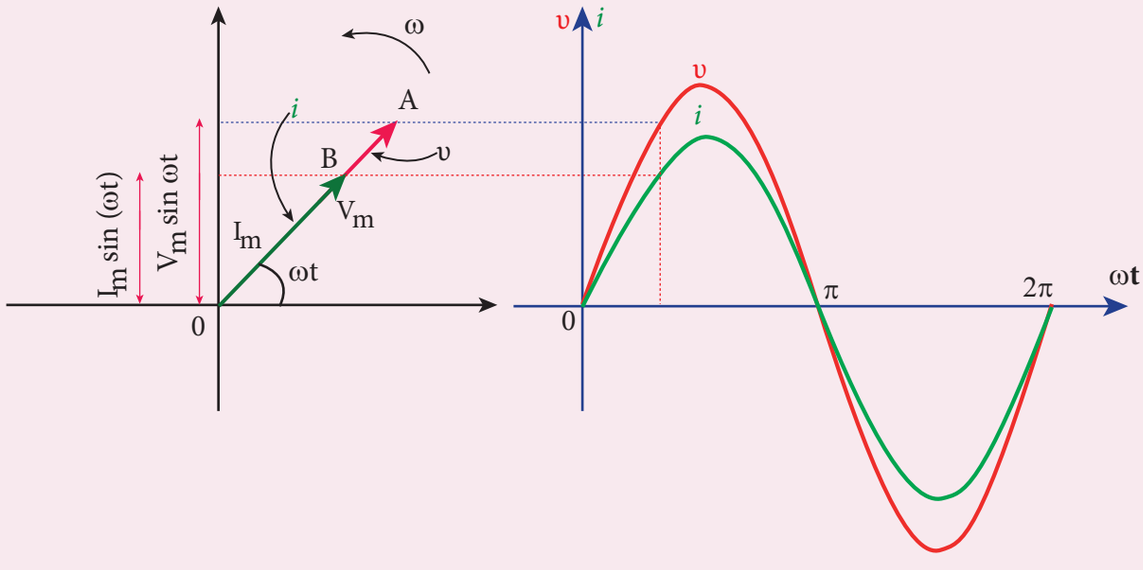


Figure 4.46 Phasor diagram and wave diagram for AC circuit with R

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